

Computer-assisted surgery for replacement of the temporomandibular joint with customized prostheses: can we validate the results?

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Replacing the temporomandibular joint poses an important challenge to maxillofacial surgeons, and for certain disorders, it represents the treatment's gold standard. Computer-assisted surgery (comprising preoperative virtual planning, virtual intraoperative navigation and 3D printing) is a useful tool for this type of surgery. However, we do not know if and how much the final position of the prosthesis differs, in absolute values, from what was planned virtually in the preoperative phase. We propose a comparative result validation system for temporomandibular joint replacement METHODS: In the present study, we propose a comparative validation system using overlapping images, between the model obtained with preoperative virtual planning and the postoperative result. The mean difference for all screws of the glenoid prosthesis was 2.08 mm (range, 1.20-3.03) and for all screws of the condylar prosthesis it was 2.33 mm (range, 1.16-3.56). Mean overall difference between both prostheses in all patients was 2.21 mm (range, 1.16-3.56). The validation system proposed by overlapping pre- and postoperative images in temporomandibular joint replacement allowed us to establish differences in absolute values between the virtual preoperative model and the actual postoperative result expressed in millimeters.

Examination of failed retrieved temporomandibular joint (TMJ) implants.

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In the management of end-stage temporomandibular joint disorders (TMD), surgeons must often resort to alloplastic temporomandibular joint (TMJ) total joint replacement (TJR) to increase mandibular function and form, as well as reduce pain. Understanding wear and failure mechanisms of TMJ TJR implants is important to their in vivo longevity. However, compared to orthopedic TJR devices, functional wear of failed TMJ TJR implants has not been examined. Not only do wear and corrosion influence TJR implant in vivo longevity, but so does reactivity of peri-implant tissue to these two events. The aim of this study was to examine and report on the wear of retrieved, failed metal-on-metal (MoM), metal-on-polymer (MoP), and titanium-nitride coated (TiN Coated) TMJ TJR implant components. A total cohort of 31 TMJ TJR devices were studied of which 28 were failed, retrieved TMJ TJRs, 3 were never implanted devices that served as controls. The mean time from implantation to removal was 7.24 years (range 3-15), SD 3.01. Optical microscopy, White Light Interferometry (WLI), Scanning Electron Microscopy (SEM), and Raman spectroscopy were utilized to characterize the surfaces of the devices. Data was acquired and evaluated by analyzing alloy microstructure. Substantial surface damage was observed between the articulating areas of the condylar head and the glenoid fossa components. Damage included pitting corrosion, evidence of deposited corrosion products, specific wear patterns, hard phases, surface depressions, and bi-directional scratches. Electrochemical analysis was performed on the MoM Control, retrieved, failed MoM, and TiN Coated devices. Electrochemical tests consisted of open circuit potential (OCP) and electrochemical impedance spectroscopy (EIS) tests conducted using the condylar head of the retrieved failed devices. EIS confirmed material properties as well as corrosion kinetics in vivo help to mitigate corrosion as reflected by the Raman spectroscopy results. In summary, this study demonstrated the role of wear and corrosion interactions on the early failure of TMJ TJR devices. Since the materials employed in most orthopedic TJR devices are similar to those used in TMJ TJR implants, studies such as this can provide data that will improve future embodiment paradigms for both. Further studies will include in vitro investigation of corrosion kinetics and the underlying tribocorrosion mechanism of TMJ TJR devices. An attempt is made in this study, to examine the retrieved TMJ implants and conduct surface and

electrochemical analysis; further a translation research approach is employed to compare the observations from the total hip replacement (THR) retrievals. A total cohort of 31 TMJ TJR devices were studied of which 28 were failed, retrieved TMJ TJRs, 3 were never implanted devices that served as controls. Data was acquired and evaluated by analyzing alloy microstructure. Substantial surface damage was observed between the articulating areas of the condylar head and the glenoid fossa components. Electrochemical analysis was performed on the MoM Control, retrieved, failed MoM, and TiN Coated devices. This study demonstrated the role of wear and corrosion interactions on the early failure of TMJ TJR devices. Since the materials employed in most orthopedic TJR devices are similar to those used in TMJ TJR implants, a comparison study was conducted.

Temporomandibular joint devices: treatment factors and outcomes.

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TMJ devices have been used for many years in reconstruction of the temporomandibular joint (TMJ). The most common endosseous implant currently used in TMJ reconstruction is a mini-anchor that is placed in the posterior head of the condyle to support artificial ligaments to stabilize the articular disk in the proper position. A 2-year follow-up study shows a success rate of 90% in reference to incisal opening jaw and occlusal stability, and significant reduction in presurgical pain level. Some materials that have been used in TMJ reconstruction, including Proplast-Teflon (PT) and Silastic devices, have caused devastating problems for patients. These materials, (particularly the PT) can cause severe foreign-body giant-cell reaction, severe bone and soft-tissue destruction, and migration of particles to other body areas, and may initiate or exacerbate connective tissue and autoimmune disease problems. Christensen joint prosthesis has been reported to have very good success in TMJ reconstruction. The most thoroughly studied TMJ total joint device is the Techmedica custom-made total joint prosthesis, with a 5-year follow-up study on 31 patients and 52 reconstructed joints. All patients have functioning prostheses with good jaw and occlusal stability and an average pain reduction of 4.4 points on a 0-to-10 visual analog pain scale. However, this device currently is unavailable. In complex cases requiring multiple TMJ operations, particularly those with previously failed alloplast, a custom-made total joint prosthesis, using materials with proven safety and efficacy in orthopedic joint reconstruction, may be the only option available to improve predictably the quality of life of these patients.

Early Clinical Outcomes of Arthroscopic Management of the Failing Alloplastic Temporomandibular Joint Prosthesis.

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Total joint replacement is the recommended treatment for end-stage temporomandibular joint (TMJ) disease. The goal of treatment is to help the return to acceptable function with improvement of the maximum incisal opening (MIO) and a reduction of pain. When a prosthetic joint shows late complications, surgical management includes an open approach, with debridement, cultures, and prosthetic replacement as options. The purpose of the present study was to evaluate the early outcomes of arthroscopic management of failing prosthetic TMJs (PTMJ). The inclusion criteria were patients with custom or stock joints with complaints of limitation of mouth opening and pain, who had undergone arthroscopy. The exclusion criteria were patients with radiographic heterotopic bone formation,

improvement with antibiotic treatment, and failed hardware found on imaging studies. A total of 9 patients were included in the present study (all women), with 5 unilateral and 4 bilateral PTMJs, for a total of 13 sides that underwent arthroscopy. Their mean age was 40 years (range, 23 to 65 years). The mean preoperative MIO was 25 mm, and the mean preoperative visual analog scale for pain and functional limitation scores were both 8 of 10. The corresponding scores were 4 of 10 and 3 of 10 at 3 months postoperatively. Arthroscopic management of prosthetic joints has been reported in orthopedic studies, with benefits shown in the diagnosis and management of synovial impingement and arthrofibrosis. The results from the present study demonstrated that the early clinical outcomes of arthroscopic management of PTMJs is promising for decreasing pain and increasing the MIO. Larger studies with longer follow-up are needed to further classify the different causes of prosthetic failure and advance the approaches to management.

Extended total temporomandibular joint replacements: a classification system.

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Prosthetic total temporomandibular joint (TMJ) replacement (TJR) is well established in the United Kingdom, with clear guidelines for indications and nationally published outcomes. CAD/CAM technology has made it possible to push the boundaries of custom-made TJR to include extended versions (eTJR), which may replace segmental mandibular defects or defects in the skull base with extended components for the ramus and fossa, respectively. Such prostheses are uncommon, and published reports are restricted to isolated cases and series of cases. We know of no previous attempts to classify such prostheses, and here we suggest a bipartite classification system for use in communications between surgeons and manufacturers based on a review of 19 prostheses provided by one manufacturer (TMJ Concepts, Ventura, CA).

[Research advances of complications regarding temporomandibular joint prosthesis replacement].

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Replacement of the temporomandibular joint (RTJ) has undergone decades of technological improvements and is now an important treatment for end-stage temporomandibular joint (TMJ) disease. The main complications of RTJ include aseptic loosening, heterotopic ossification, postsurgical infection, hypersensitivity reaction, prosthesis loosening or malposition, nerve injury and severe bleeding. Improvement in clinical technique is now the key to resolving complications. In the future, improvements in prosthetic materials may be an important development for newer iterations. With the development and popularity of TMJ surgical techniques, prosthesis-related complications will increasingly come into clinical view. This article provides a systematic review of the complications of RTJ prosthesis replacement and the measures to deal with them, and looks forward to the direction of the development of this field from the perspective of reducing complications, so as to provide a reference for clinical work.

Inebilizumab for Treatment of IgG4-Related Disease.

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IgG4-related disease is a multiorgan, relapsing, fibroinflammatory, immune-mediated disorder with no approved therapy. Inebilizumab targets and depletes CD19+ B cells and may be effective for treating patients with IgG4-related disease. In this phase 3, multicenter, double-blind, randomized, placebo-controlled trial, adults with active IgG4-related disease underwent randomization in a 1:1 ratio to receive inebilizumab (300-mg intravenous infusions on days 1 and 15 and week 26) or placebo for a 52-week treatment period. Participants in both groups received identical glucocorticoid tapers. Glucocorticoids were allowed to treat disease flares, but background immunosuppressants were not permitted. The primary end point was the first treated, adjudicated disease flare during the treatment period, assessed in a time-to-event analysis. Key secondary end points were the annualized flare rate and treatment-free and glucocorticoid-free complete remission. A total of 135 participants with IgG4-related disease underwent randomization: 68 participants were assigned to receive inebilizumab and 67 were assigned to receive placebo. Treatment with inebilizumab reduced flare risk; 7 participants (10%) in the inebilizumab group had at least one flare, as compared with 40 participants (60%) in the placebo group (hazard ratio, 0.13; 95% confidence interval [CI], 0.06 to 0.28; $P < 0.001$). The annualized flare rate was lower with inebilizumab than with placebo (rate ratio, 0.14; 95% CI, 0.06 to 0.31; $P < 0.001$). More participants in the inebilizumab group than in the placebo group had flare-free, treatment-free complete remission (odds ratio, 4.68; 95% CI, 2.21 to 9.91; $P < 0.001$) and flare-free, glucocorticoid-free complete remission (odds ratio, 4.96; 95% CI, 2.34 to 10.52; $P < 0.001$). Serious adverse events occurred during the treatment period in 12 of the participants (18%) who received inebilizumab and 6 of the participants (9%) who received placebo. Inebilizumab reduced the risk of flares of IgG4-related disease and increased the likelihood of flare-free complete remission at 1 year, confirming the role of CD19-targeted B-cell depletion as a potential treatment for IgG4-related disease. (Funded by Amgen; MITIGATE ClinicalTrials.gov number, NCT04540497.).

Efficacy and Safety of Inebilizumab in IgG4-Related Disease: Protocol for a Randomized Controlled Trial.

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Immunoglobulin G4-related disease (IgG4-RD) is a debilitating multiorgan disease characterized by recurring flares leading to organ dysfunction, decreased quality of life, and mortality. Glucocorticoids, the standard of care for IgG4-RD, are associated with substantial treatment-related toxicity. Inebilizumab, an antibody directed against CD19, mediates the rapid and durable depletion of CD19+ B cells thought to be involved in IgG4-RD pathogenesis. We describe the first international, prospective, double-blind, placebo-controlled trial to evaluate the safety and efficacy of B-cell depletion for flare prevention in IgG4-RD (MITIGATE). The study was designed by an international panel of physicians with expertise in IgG4-RD. Critical trial design decisions included the selection of participants, definition of clinically meaningful primary and secondary endpoints, accommodation of standard of care, and development of flare diagnostic criteria. The study is approved for conduct in 22 countries. The primary efficacy endpoint is time from randomization to the occurrence of the first centrally adjudicated and investigator-treated disease flare during the 1-year randomized controlled period. A set of novel, organ-specific flare diagnostic criteria were developed specifically for this trial, incorporating symptoms and signs, laboratory findings, imaging study results, and pathology data. MITIGATE aims to accrue 39 flares for the primary endpoint, which provides sufficient power to detect a relative risk reduction of 65% in the inebilizumab group. It is anticipated that enrollment of 160 participants will achieve this goal. Additional endpoints

include safety, annualized flare rate, flare-free complete remission, quality-of-life measures, and cumulative glucocorticoid use. MITIGATE represents the first randomized, double-blind, placebo-controlled trial of any treatment strategy conducted in IgG4-RD. Data from this study will provide insights into the natural history and pathophysiology of IgG4-RD and the efficacy and safety of B-cell depletion as a therapeutic avenue. NCT04540497.

Inebilizumab: First Approval.

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Inebilizumab (Uplizna™; inebilizumab-cdon in the USA) is a humanised anti-CD19 monoclonal antibody being developed by Viela Bio for the treatment of a range of autoimmune diseases associated with CD19-expressing B cells. Inebilizumab targets and depletes CD19-expressing B cells through antibody-dependent cell-mediated cytotoxicity. In June 2020, inebilizumab received its first global approval in the USA for the treatment of neuromyelitis optica spectrum disorder (NMOSD) in adult patients who are seropositive for immunoglobulin G autoantibodies against aquaporin-4 (AQP4-IgG). The drug is also undergoing clinical evaluation for kidney transplant desensitization, myasthenia gravis, and IgG4-related disease. This article summarizes the milestones in the development of inebilizumab leading to this first approval for the treatment of AQP4-IgG seropositive NMOSD.

New Developments in Anti-Reflux Surgery: Where Are We Now?

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Gastroesophageal reflux disease is one of the most common chronic diseases, affecting up to 28% of the western population. Therapeutic management ranges from conservative measures to endoscopic or surgical interventions. Laparoscopic Nissen fundoplication (LNF) still is considered as gold standard, but alternative procedures have been developed and evaluated within the past years. Magnetic sphincter augmentation (MSA), which aims to be a less disruptive and possibly more standardized laparoscopic procedure than LNF, shows satisfying results regarding short- and long-term follow-up as well as comorbidities. Alternatives, such as the RefluxStop™ procedure or Transoral incisionless fundoplication (TIF), combined with laparoscopic hiatal hernia repair (cTIF) show promising results for short-term follow-up, nevertheless further studies regarding long-term follow-up are necessary. Although there definitely are upcoming and promising trends in upper GI surgery, LNF still represents today's gold standard and MSA is an equivalent alternative. RefluxStop™ and cTIF show promising results, nevertheless, further studies are necessary.